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# Qualitative Analysis of Visualization: A Building Design Field Study

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## ABSTRACT
We conducted an ethnographic field study examining the ways in which building design teams used visual representations of data to coordinate their work. Here we describe our experience with this field study approach, including both quantitative and qualitative analysis of field study data. Conducting a field study enabled us to effectively examine real work practice of a diverse team of experts, which would have been nearly impossible in a laboratory study. We also found that structured qualitative analysis methods provided deeper insight into our results than our initial quantitative approach. Our experience suggests that field studies and qualitative analysis could have substantial benefit in visualization and could nicely complement existing quantitative laboratory studies.

## Categories and Subject Descriptors
H.5.2 [**User Interfaces**]: Evaluation/Methodology; H.1.2 [**User/Machine Systems**]: Human Factors; H.5.3 [**Group and Organization Interfaces**]: Evaluation/Methodology; J.5 [**Arts and Humanities**]: Architecture; J.6 [**Computer-Aided Engineering**]: Computer-aided design (CAD).

## General Terms
Experimentation, Human Factors.

## Keywords
Visualization, Evaluation, Qualitative analysis, Ethnographic field study, Long-term study.

## 1. INTRODUCTION
Evaluation of visualization tools and techniques is a growing trend. Several years ago, few visualization systems were evaluated at all. By contrast, most current systems are evaluated in some way, often via user studies. User studies can be classified very broadly by their location: in the *field* or in the *laboratory*, and by their type of analysis: *qualitative* and/or *quantitative*. By and large, most formal user evaluations of visualization systems have involved quantitative laboratory experiments. In these studies, users are asked to complete a series of benchmark tasks with two or more visualization systems or methods, with time and error rates as the primary metrics. Many examples of such studies exist in the literature (e.g.,[7][13]), including several of our own (e.g., [21][22]). This reductionist approach has been adapted from other fields such as psychology, and follows good scientific practice: control extraneous factors, isolate the variable of interest, and demonstrate results statistically. Shneiderman and Plaisant [17] provide a useful review of how these methods have evolved through history. We support and encourage quantitative experiments and believe they provide some valuable knowledge about visualization design. But we also believe they miss something important.

Consider Kobsa's comparison of commercial information visualization systems [7] that revealed which systems were fastest and most accurate for various benchmark tasks. Because the systems differed from each other in many ways, including visual design, interaction style, and available visualization techniques, the author could only guess at the reasons behind performance differences. Laboratory studies generally test novice users conducting simplistic tasks on arbitrary data for a short time period. Benchmark tasks are usually simplistic so that they can be carefully controlled and accuracy can be explicitly measured. Unfortunately, this approach limits the realism of the studies. If the goal of visualization is for *expert users* to gain *insight* into their *own data* and *communicate it to others* over an *extended time* period, and the goal of evaluation is to *understand how to best design visualization tools* to support these users, then these quantitative lab studies clearly fall short. Ideally, we would like to understand which features of a visualization tool make it useful for complex, long-term data analysis by domain experts. Alternate evaluation methods are needed to support these goals.

We believe field studies and qualitative analysis can complement quantitative laboratory studies to provide this missing understanding. In this paper, we discuss our experiences with both quantitative and qualitative analysis of a field study, and describe how the qualitative analysis led to a much richer understanding. In the study, we focused on visual representations used by a building design team during collaborative meetings. Detailed results of the study will be reported elsewhere (e.g. [23]). In this paper, we focus on the effectiveness of our methods, with some example results to illustrate our points. We discuss the benefits and

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challenges of qualitative methods and field studies, and suggest how they might be further applied in the field of visualization.

In the remaining sections, we first discuss related visualization studies involving qualitative analysis and/or field work. We then describe our field study that examined the use of visual representations by a building design team. We focus first on our initial quantitative analysis and then on our later qualitative approach. We end with a discussion of lessons learned from this experience and conclusions about the appropriateness of these methods for evaluating visualizations.

## 2. RELATED WORK

Several recent papers have stressed the need for new methods of user testing. These papers have argued for evaluating visualization techniques in more realistic situations [14][17], measuring complex outcomes such as 'insight' [15][16], and understanding the design features that contribute to a visualization technique's success or failure for various tasks [15]. Many of these discussions point towards longitudinal field studies and / or qualitative analysis; however, actual experience with these evaluation methods for visualization is limited.

Some quantitative laboratory experiments have incorporated a qualitative aspect. For example, Kobsa [7] attempted to understand reasons behind performance differences when using different systems based on qualitative observations. Similarly, Li & North [9] used participant comments to explain some of their quantitative results, such as a misunderstanding of the meaning of histogram bars by some participants. In such studies, observations and user comments help to explain the statistical results, but the qualitative analysis is usually limited in depth and secondary to the quantitative one. Others have collected qualitative data through interviews and focus groups [11][15]. Rester [15] recommends using a mixture of evaluation methods since each will provide a different perspective.

Kobsa [8] formalized the qualitative aspect of a visualization experiment for the comparison of tree visualization systems. A structured qualitative video analysis was performed in addition to a quantitative comparison. He reported that the qualitative analysis provided insight into usability problems and revealed the reasons why some systems performed better than others. Using video analysis rather than live note taking enabled the experimenters to use two independent observers who could go back to the video independently or together to review and update their findings. Similarly, Tang et al. [19] used an open-coding approach to video analysis to identify collaboration strategies with map and network visualizations. Our current study uses similar qualitative video analysis methods, but with a focus on experts' current work practice over an extended time period rather than short evaluation sessions with novice users.

Longitudinal use by expert users is increasingly recognized as an important aspect of evaluation [14][16][17]. Shneiderman and Plaisant [17] introduced the concept of multi-dimensional in-depth long-term case studies (MILCs), where a small group of users are studied in detail, using multiple evaluation methods, over an extended time period of weeks or months. The MILC approach builds upon a long history of qualitative methods. These originated in the social sciences and have been more recently adopted in HCI. For a general introduction, see [3]. Our study follows in this methodological tradition, but differs somewhat from MILCs. In particular, our focus was on understanding current work practice with visual representations rather than evaluating a new visualization technique, similar to Faisal et al.'s [4] study of how researchers used knowledge domain visualizations to examine literature in their field. As such, we did not have specific hypotheses at the beginning of the study and our work was more ethnographic in nature. In addition, Shneiderman and Plaisant [17] provide many suggestions for conducting long-term field studies, but limited guidance on how to analyze the resulting data. Here we describe our experiences with both quantitative and qualitative analysis, with the hope that that they will provide guidance for future studies.

## 3. FIELD STUDY OF BUILDING DESIGN COORDINATION

Building design is a complex process involving vast amounts of information shared by a diverse team of stakeholders. Much of the information is represented visually. This includes representations of the design itself (CAD drawings, digital and physical design models, energy models, etc.) as well as representations that support the design process (schedules, cost information, etc.). Decisions made during building design involve and affect many stakeholders, including architects, engineering consultants, construction managers, facility maintenance organizations, facility users, and property managers. For example, a seemingly simple change in the ceiling height of a floor could affect the space available for mechanical and electrical equipment, lighting requirements (because of changes in the amount of daylight entering the room through exterior windows), and the construction cost. Coordination among stakeholders, therefore, is critical to ensure that the building design meets the functional, aesthetic, and economic requirements of the owner. Visual representations play a critical role in mediating this coordination.

**Figure 1: Typical design coordination meeting for a building design project. Participants are examining a paper-based 2D CAD drawing representing part of the building design.**

Interactive visualization tools play an important role in the building design process. Individual design work is usually accomplished using two-dimensional (2D) and three-dimensional (3D) computer aided design (CAD) software. Schedule information is often represented using Gantt and Pert charts created in software such as Microsoft Project. However, coordination among stakeholders during in-person meetings often involves much lower-tech visual representations such as paper

prints of drawings, schedules, and spreadsheets, as shown in Figure 1. A growing trend in building design is to model the building and its systems entirely in 3D CAD. A 3D design process has several known advantages, including fewer design conflicts and errors [18], improved data access and integration [2], and enhanced visualization and interaction [12]. However, 3D models cannot be easily printed, requiring a change in the way design teams meet to discuss and coordinate their work. It remains unclear how to best integrate digital technology and interactive visualization tools into design coordination meeting situations. This was the focus of our study.

## 3.1 Study Objectives

Our main goal was to understand how visual representations are currently used during coordination meetings, and to use this information to identify design guidelines for interactive visualization tools that would be used during meetings. We also wanted to compare how digital and physical representations of data were used. Toward this end, we examined two elements of the meeting process in parallel:

- *Interactions*: Low level tasks and interactions involving visual representations of data. For example, users might search or query a CAD drawing for information, and interact with drawings by pointing and turning pages to see different views.
- *Meeting Process*: The higher level meeting process and productivity, including artifact utilization and the role of visual representations. For example, meeting discussions covered a flow of different kinds of meeting tasks and meeting participants utilized different types of artifacts in their decision-making process.

## 3.2 Data Collection and Analysis

Because our goal was to understand current work practices in a real setting, a laboratory study would have imposed too many artificial constraints. Instead, we conducted an ethnographic field study where we observed coordination meetings of a particular building design team as they transitioned from paper-based meetings to digital meetings. We also conducted interviews with key members of the team.

In total, we observed 20 in-person coordination meetings over a seven-month period. Each meeting was approximately 3 hours long. Early meetings were conducted in a traditional boardroom (*paper-based meetings*) and later meetings were conducted in a digital meeting space containing two large screen SmartBoard displays (*digital meetings*). Two types of meetings were observed: *design development meetings* that focused on refining the design of various systems in the building, and *value engineering meetings* that focused on reducing the cost of the building project.

We interviewed the most frequent meeting participants: the architects, construction manager, electrical consultant, and mechanical consultant. The second author actively participated in the building project as a 3D design consultant and attended several meetings in this role. The other author acted as a passive observer during meetings. Meetings were videotaped and interviews were audio-recorded. We also had access to a document-sharing system (Autodesk Buzzsaw [1]), which the building design team used to share and archive documents relevant to the meetings, including CAD drawings, CAD models, agendas, meeting minutes, consultant reports, schedules, and cost plans. Only part of this data was collected at first; interviews were conducted and several meeting videos were analyzed only after the initial quantitative analysis.

### *3.2.1 Initial Analysis: Quantitative*

**Quantitative Analysis Method**

Our initial approach to analyzing the data was primarily quantitative, where we counted and timed events of different types using video coding. Our intent was to quantitatively 'characterize' the meetings and provide measures by which we could compare various aspects of meetings. For example, we wanted to measure meeting productivity and then compare whether it was higher when digital representations of data were present.

*Interactions:* To study low level tasks and interactions, we counted various interactions with artifacts (paper diagrams, models, electronic devices, etc.) using quantitative video coding. Our intent was to capture work patterns and interactions with various artifacts. Each time a meeting participant referred to or interacted with an artifact, we recorded the artifact in question, the action performed, the perceived goal of performing the action, and whether the action was part of a group, subgroup, or individual activity. We selected artifacts, actions, and goals for our coding scheme based on ad-hoc qualitative observations at the meetings. A sampling of this coding scheme is given in Table 1. Eight meeting videos were analyzed in this way.

**Table 1: Partial video coding scheme for quantitative analysis of low-level interactions with artifacts.**

| ***Artifacts*** | ***Actions*** | ***Goals*** | ***Scope*** |
|---|---|---|---|
| Paper Diagram | Manipulate | Explain | Group |
| 3D model | Point / Gesture | Gain attention | Subgroup |
| Paper | Compare | Indicate | Individual |
| Pen | Move / Rotate | Lookup | |
| Notepad | Pass artifact | Decide | |
| Laptop | Annotate | Make query | |
| PDA | Browse | Share | |
| White board | Sketch | Prepare | |
| etc… | etc… | etc… | |

*Meeting Process:* Similarly, to examine the higher-level meeting process and productivity, we employed a coding scheme and measures defined by Liston et al. [10]. Specifically, we categorized the type of discussion tasks observed throughout the meetings, as in the second column of **Table 2**. We then used the relative percentages of time spent on these tasks to calculate measures of meeting productivity as a function of task resolution (e.g., Equation 1), as suggested by Liston. We also kept track of information types and forms used in different meetings (see **Table 2**). We used these to identify which types of information were used when meetings were more or less productive and to get a sense for the information complexity. Eight meeting videos were analyzed in this way.

**Table 2: Video coding scheme for quantitative analysis of the meeting process.**

| ***Subjects*** | ***Tasks*** | ***Information Type*** | ***Information Form*** |
|---|---|---|---|
| Architect | Descriptive | Temporal Chart | Temporal |
| Structural Engineer | Explanative | 2D | Spatial |
| Mechanical Engineer | Evaluative | 3D Physical Model | Quantitative |
| Electrical Engineer | Predictive | 3D Virtual Model | Symbolic |
| Landscape Architect | | 4D | Semantic |
| Code Consultant | | Text | |
| Project Manager | | Diagram | |
| Construction Manager | | Symbol | |
| Mechanical Contractor | | Chart | |
| Electrical Contractor | | Table | |

(1)

$$meeting\ \ productivity = Average\left(\frac{\sum_i Duration\ \ of\ resolved\ \ tasks}{\sum_i Duration\ \ of\ total\ \ tasks}, \frac{\sum_i No.\ of\ resolved\ \ tasks}{\sum_i No.\ of\ total\ \ tasks}\right)_{meeting}$$

**Quantitative Analysis Results**

The quantitative analysis led to some interesting observations. For instance:

- Meeting participants spent a significant amount of time on descriptive (35%) and explanative (42%) tasks and very little time on predictive (12%) and evaluative (11%) tasks.
- Meeting productivity was quite high, with an average of 73%.
- Meeting environment substantially changed the distribution of low level interactions with visual representations. Pointing and gesturing accounted for 84% of interactions in paper-based meetings and 47% in digital meetings. Manipulating and accessing accounted for 8% of interactions in paper-based meetings and 29% in digital meetings.
- Meeting content (design development vs. value engineering) had minimal impact on the distribution of low level interactions with visual design representations.

However, despite these interesting observations, our quantitative analysis left us with many unanswered questions and inconclusive results. For example, we questioned the validity of the quantitative measures of meeting productivity. How do you define meeting productivity considering the variety of meeting goals and outcomes? If a task is resolved, does that necessarily mean it was a worthwhile task? We were also not sure how to interpret our characterization of meeting tasks. What does it mean if 47% of the time in a particular meeting was spent on descriptive tasks? Is that good or bad? Similarly, what does it mean if the types of interactions with information are different in different meeting environments? Does this matter?

In our initial analysis, we also made several interesting qualitative observations. For example:

- Contrary to our expectations, the team rarely annotated design drawings during meetings. Instead, they sometimes made explicit drawing-like actions in the air over the page or sketched on blank paper layered over the drawing.
- The architect was able to 'bookmark' pages in a stack of paper drawings by either placing his hand at a location to mark it, by pulling a page partially out of the stack, or by holding a page in the air. We believed that these could be used as metaphors for digital bookmarking.
- An architect occasionally pointed at paper drawings using all five fingers to surround an area of interest. This suggested a possible input technique where *five-finger pointing* could be used to control a spotlight that would highlight information.
- We observed that meeting participants did not utilize artifacts very effectively to communicate their designs, although gesturing to design drawings was very common.

Many of these qualitative observations were interesting. However, the observations were very ad-hoc and left many open questions. Reasons behind many of our observations were unknown. For example, what was the reason why drawings were rarely annotated? Were there other methods of bookmarking and other distinctive pointing styles that we had not noticed? In addition, these observations were often based on one or two instances and individuals. It was not at all clear whether these actions were consistent across people, and whether they would occur frequently or were simply one-time events.

### *3.2.2 Later Analysis: Qualitative*

Following our initial quantitative analysis, we conducted an in-depth qualitative analysis to address the same research questions.

**Qualitative Analysis Method**

We took an approach inspired by grounded theory [5]. The goal of a grounded theory study is to generate a theory by systematically analyzing 'grounded' (i.e. real-world) qualitative data. The analysis typically begins with open-coding as described below. Additional data is collected and analyzed when the current data seems insufficient. Our analysis primarily centered on videos. Interviews were used to validate findings and identify issues that participants felt were critical in reference to integrating digital media into work practice.

We used an open-coding method to systematically analyze the video data, similar to other research [6][20]. Open-coding is a standard method for extracting themes and theories from a body of qualitative data. We focus here on video data, but the same method can be applied to transcribed interviews. As the researchers watch the videos, they take notes of interesting occurrences and create keywords or *codes* to describe those events. Similar events are assigned the same code. Codes are iteratively refined, added, deleted, and categorized into groups as the analysis progresses; relationships between codes are also iteratively identified and refined. Early in the analysis, any interesting event might be given a code, but subsequently, codes are considered valid only if the event occurs in multiple video segments. Researchers re-watch sections of the video multiple times (e.g. to compare whether events are similar enough to be

**Table 3: Iteratively refined coding scheme for qualitative analysis of the meeting process.**

| ***Subjects*** | ***Scene*** | ***Tasks*** | ***Artifact*** | ***Artifact Type*** | ***Artifact Form*** | ***Bottlenecks*** |
|---|---|---|---|---|---|---|
| Architect | Design Details | Descriptive | Stick Set | Temporal Chart | Temporal | Data Access |
| Structural Engineer | Budget Items | Explanative | Cost Plans | 2D | Spatial | Visualization |
| Mechanical Engineer | Exterior Design | Evaluative | Personal Notes | 3D Physical Model | Quantitative | Collaboration |
| Electrical Engineer | Equipment location | Predictive | Sketches | 3D Virtual Model | Symbolic | Research |
| Landscape Architect | Government regulations | | | 4D | Semantic | Negotiation |
| Code Consultant | Glazing | | | Text | | |
| Project Manager | Water and wastewater | | | Diagram | | |
| Construction Manager | etc... | | | Symbol | | |
| Mechanical Contractor | | | | Chart | | |
| Electrical Contractor | | | | Table | | |

given the same code, to determine whether events in a group should be divided into subcategories, or to search earlier segments for instances of codes that were identified while watching later segments). The research questions themselves also iteratively change throughout the analysis as interesting observations, trends, and relationships emerge.

*Interactions*: To examine low level interactions, we analyzed ten meetings in depth. We iteratively characterized the ways in which the design team interacted with design information, and the goals participants were trying to achieve by means of the interactions. *Codes* were names for commonly observed behaviours or phenomena of interest, such as *bookmarking* and *five-finger pointing* as described in the previous section. We also noted bottlenecks in the meeting process that could be attributed to awkward or time-consuming interactions with artifacts. We focused on physical interactions with design information, but used participants' speech to understand the intent and purpose of those interactions, similar to Tang [20].

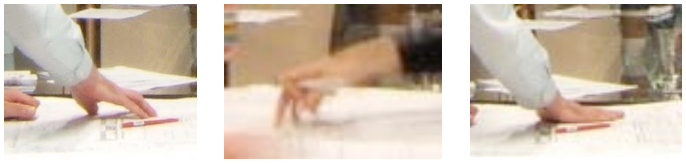
*Meeting Process*: To examine the high level meeting process, we examined six meetings in detail and iteratively characterized the content of the meeting discussion and the meeting tasks (coded previously) as in Table 3. Then, for each meeting task, we identified the artifacts utilized and characterized the bottlenecks encountered. We characterized the meeting content as *scenes* that illustrate the general meeting topics and involve one or more meeting tasks. Artifacts characterized the different domain-specific documents utilized in the meeting task. Bottlenecks were encountered when meeting workflow was disrupted or when the completion of meeting tasks was impeded or inefficient. We characterized the type of bottleneck to understand how different digital technologies might help to overcome it. For example, confusion around the type of wall and the size of an air conditioning unit in one meeting suggested that the type of visual representation used limited participant understanding. This approach involved viewing the video repeatedly to capture the essence of the meeting discussion and the nature of the bottlenecks encountered.

**Qualitative Analysis Results**

*Interactions*

Our analysis of low level interactions resulted in a detailed taxonomy that explained how different representations of design information were used to facilitate collaboration during coordination meetings. It also revealed many interaction bottlenecks that reduced productivity. A complete description of these results is reported elsewhere [23]. Here we describe some example results to demonstrate that the qualitative analysis led to a richer understanding of the interactions participants had with visual representations. The analysis also clarified the relative importance of our observations.

Our initial observation of five-finger pointing encouraged us to further examine pointing and gesture interactions related to design information such as drawings. Our initial coding scheme included simply five-finger pointing and ordinary one-finger pointing. By the time we finished the qualitative analysis, this had been expanded to nine distinct categories of pointing based on hand shape and movement. A few of these are shown in Figure 2. We also learned that these categories were consistent across individuals with varied backgrounds, different types of meeting content, and different types of visual representations including digital drawings, paper-based drawings, and physical models. This consistency suggests that these pointing techniques could be used as intuitive and natural methods of input.



(a) (b) (c)

**Figure 2: Example pointing techniques. (a) *five-finger pointing* where five fingers surround the area of interest (b) *two-finger indication* where two fingers indicate the space between (c) *covering* where the hand covers the area of interest**

Similarly, our initial observation of paper-based bookmarking encouraged us to examine bookmarking in our qualitative analysis. The new analysis led to the discovery of an additional method for bookmarking: rolling groups of pages away from a stack. We also observed many more instances of these bookmarking methods by multiple people, providing confidence that these were not isolated one-time events.

More importantly, examining bookmarking led us to study the higher-level category of *navigation* – changing which information was visible or altering one or more participants' view of the information. We developed a detailed taxonomy of navigation mechanisms for visual design information. We also realized that navigating digital drawings was a major bottleneck in meetings, taking substantially more time than navigating the equivalent paper drawings. We concluded that 3D representations may have been helpful in some situations, and that better integration between 2D and 3D views was needed to improve usability. We also suggested developing a specialized application for viewing digital 2D CAD drawings during meetings. Such a viewer would include features such as an overview of the collection of drawings and easily customizable shortcuts to move quickly back and forth between drawings or parts of drawings. Such features are common practice in visualization. However, prior to this study, we would not have believed that they were necessary for navigating within a small collection of approximately ten non-interactive drawings and that this would be a key bottleneck in the meeting process.

*Meeting Process*

Our initial reservations about the meeting productivity rates and the vagueness of the meeting task types led us to pursue a more holistic approach to analyzing the high level meeting process. We wanted to gain a better understanding of the content of the meeting discussion and the factors that contributed to meeting effectiveness. Therefore, our qualitative analysis shifted the focus from an emphasis on tasks to an emphasis on scenes and from an emphasis on productivity to an emphasis on bottlenecks.

The results of our qualitative analysis of the meeting process is a detailed characterization of the meeting context, which includes scenes, the type and flow of meeting tasks, and the bottlenecks encountered. We also classified the artifacts utilized for different meeting tasks and in certain cases, calculated the artifact utilization rate for the different meeting environments. A detailed description of these results is in preparation.

We extended the coding scheme to include *scenes* and domain-specific *artifacts* to get a better sense for the generality of the meeting tasks and whether they impacted the distribution and flow of meeting tasks (see Table 3). For example, we analyzed discussions of the equipment location, exterior design, and cost plan to determine if one type of discussion topic required more description or explanation than other topics. We found that the discussion topics had little impact on meeting task distribution and flow. These conclusions gave us more confidence in the generality of the meeting task types.

We analyzed the *bottlenecks*, *artifacts*, and *artifact types* to get a better sense for the causes of productivity problems. We found that the types of visual representations used and the limited interactivity of those representations led to significant bottlenecks. We also found that data access was a significant problem even when the artifacts were available, though required some effort to obtain in the meeting setting. We found this characterization to be a much more useful way of critiquing meeting productivity and identifying design guidelines for digital tools to support meeting activities.

# 4. LESSONS LEARNED

## 4.1 Field Studies

Our field study allowed us to examine how visual representations were used by a group of building design experts on a real project. Because our research questions involved understanding real work practice, they would have been difficult to accurately answer in a laboratory study, where conditions are controlled and do not accurately represent the messiness of real practice. Realism is the greatest advantage of field studies. We believe the opportunity to observe a diverse team of experts doing real design work was well worth the challenges we encountered.

We made a number of decisions in the design of our study that affect the generality of our results. We chose to study one group in depth rather than multiple groups because this would better enable us to gain a deep understanding of a team's work practice, something that would not be practical in a study with greater breadth. However, it is unclear whether our results can be directly generalized from the design team and project that we studied. Further observation of other design teams is needed to corroborate our own evidence, but would be very time consuming to accomplish. Data collection even for one design team took time; the team we studied spent a substantial amount of time discussing issues that had little relevance to our research questions, and the study had to be conducted over seven months to correspond with the building design schedule.

Another disadvantage of field studies is the limited technical quality of data collected. Because we could not set up the meeting environment, we were limited in where we could place video cameras and microphones. Ideally, we would have liked several cameras from different viewpoints, including one looking down at the meeting table where paper drawings were placed; however, this was not practical to set up for each meeting. The equipment also needed to be portable so we could easily bring it to and from the field site. In the end, we used one video camera and attempted to find the best placement to view the meeting scene. Often however, the placement and limited resolution of the camera did not allow us to see details such as specific items participants pointed to in paper or digital drawings.

## 4.2 Qualitative Analysis

The key lesson we learned from this study was the effectiveness of qualitative analysis. Our initial approach focused heavily on quantitative or 'hard' evidence. Some of our numeric results were useful, but we also questioned the validity and meaning of many of our results. By contrast, our qualitative results provided a rich,

thorough understanding of the complex processes involved in collaborative design meetings.

An important aspect of qualitative analysis is *triangulation*, or collecting information about each research question from different sources. Initially, we focused our analysis almost entirely on video data. Later, by supplementing this information with interviews of various participants we gained more confidence in our conclusions. For example, participants could explain the reasons behind actions that we had observed them doing. We could also compare participants' impressions about meeting bottlenecks to the bottlenecks we observed. The importance of triangulation has also been reported by others [14][15].

We also found that structuring the qualitative analysis was very important. Our initial ad-hoc observations (e.g., bookmarking and five-finger pointing) were limited in their scope and usefulness, and their validity was questionable since they were based on only one or two observations. Although the structured open-coding method took a substantial amount of time, the thorough analysis led to more in-depth results with greater validity. A theory about current work practice emerged and observations fit together into a higher-level structure to provide a thorough understanding.

As scientists and engineers trained in reductionist scientific thinking, adopting qualitative methods requires overcoming some deeply ingrained assumptions. First and foremost, research questions and hypotheses cannot be precisely articulated at the beginning of the study. Rather, results emerge gradually and research questions evolve as the study proceeds. It is not possible to predict exactly which data will need to be collected or when data collection will be complete. In addition, the validity of research is not determined by whether or not it is backed up by numeric evidence. Structured qualitative analysis can be equally as valid as quantitative analysis. However, it is important to distinguish between qualitative analysis that is ad hoc and qualitative analysis that follows structured methods of inquiry.

# 5. EXTENSIONS TO OTHER STUDIES

This section discusses how the methods we used might be applied to other application areas and to evaluate new visualization tools and techniques.

## 5.1 Understanding practice

Our study aimed to understand and build a theory about current work practice. The methods we used are well suited to this purpose, and would likely work for many different applications. For example, we focused on a collaborative building design task, but work practices of a single biological analyst could be characterized equally well. Researchers could observe and videotape the analyst at their workplace (e.g., when working with biological visualization tools and other software), and interview the analyst about how they work. Video and interview data could be analyzed using a similar open-coding approach to ours, in order to identify strategies and bottlenecks in the work process. These results could be used to design new visualization techniques.

## 5.2 Evaluating tools

We believe field studies and qualitative analysis would also be useful for evaluating new visualization tools and techniques, although this type of study would be somewhat different from our own. Unlike typical laboratory studies, deploying a visualization tool in the field could allow researchers to examine use of the tool over an extended time period for a real application. One downside of this approach is that the tool would likely need to be more refined than for a short quantitative study, possibly including extra features needed for practical long-term use. There are a variety of qualitative research traditions, including grounded theory, phenomenology, case studies, etc. [3]. We took an approach based on grounded theory [5], which is useful for understanding current practices and developing theories. Evaluating visualization tools by deploying them in the field is more of a case study approach, but could use a similar method of structured qualitative analysis.

Qualitative analysis is not limited to field studies; it could be applied equally well to structured laboratory studies. This approach could be useful for comparing two or more tools or techniques in a short time frame. Video data could be collected as a record of how people interacted with the tool and interviews could be conducted to obtain users' impressions. Qualitative analysis of this data could be particularly useful for studying complex, open-ended tasks where performance cannot be easily measured quantitatively.

Regardless of whether qualitative data is collected in the field or in the lab, structured qualitative analysis methods could be used to gain a holistic understanding of a visualization tool's effectiveness. An open-coding approach similar to ours could be used to identify bottlenecks in working with the tool, strategies taken by users to accomplish their goals, ways in which users work around bottlenecks, and the suitability of different work strategies for different visualization techniques.

We suggest that the visualization community should adopt qualitative analysis methods and conduct more qualitative studies to complement the quantitative ones currently being done. These qualitative studies will be most beneficial if researchers apply qualitative analysis techniques in a structured and rigorous way, and if publication venues recognize the value of rigorous qualitative analysis and the substantial time it requires.

# 6. CONCLUSIONS

Field studies and qualitative analysis can provide depth of insight into visualization design that is not possible with strictly controlled quantitative experiments. Qualitative analysis enables us to gain a thorough understanding of the complex benefits and drawbacks of a visualization, including diverse aspects such as the design itself, integration with other analysis tools, impacts on work practice, and so on. Similarly, long-term field studies provide an in-depth look at real use of visualization tools by actual users. These methods cannot provide 'proof' that one visualization technique is better than another. However, such quantitative proof is often highly limited in its applicability to real practice because of constraints imposed on the interfaces, tasks, and length of evaluation. Although qualitative analysis and field studies are not new, they have been rarely employed in visualization. We suggest that using these methods more widely, to complement quantitative studies, will greatly enhance our collective understanding of effective visualization design.

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